Question 3 - Automate application deployment

Question 3: Automate application deployment using Jenkins by configuring build triggers, post-build actions, and deployment steps.

Repository: https://github.com/ksamirdev/ensp461-devops-lab1

Build trigger

The Jenkinsfile declares pollSCM('H/2 * * * *'), so Jenkins checks GitHub about every two minutes.

```
$ git diff
diff --git a/service/web.py b/service/web.py
index e1ce503..07e2eb6 100644
--- a/service/web.py
+++ b/service/web.py
@@ -3,7 +3,7 @@ import os

 from flask import Flask, jsonify, request

-RELEASE = os.environ.get("RELEASE", "1.0.0")
+RELEASE = os.environ.get("RELEASE", "1.1.0")
 app = Flask(__name__)



$ git add service/web.py

$ git commit -m "Bump release to 1.1.0 to exercise automatic deployment"
[main e93d90f] Bump release to 1.1.0 to exercise automatic deployment
 1 file changed, 1 insertion(+), 1 deletion(-)

$ git tag -a v1.1.0 -m "Release deployed by SCM polling"
```

Fig: Release change committed and tagged (rendered from evidence/logs/q3-trigger.log)

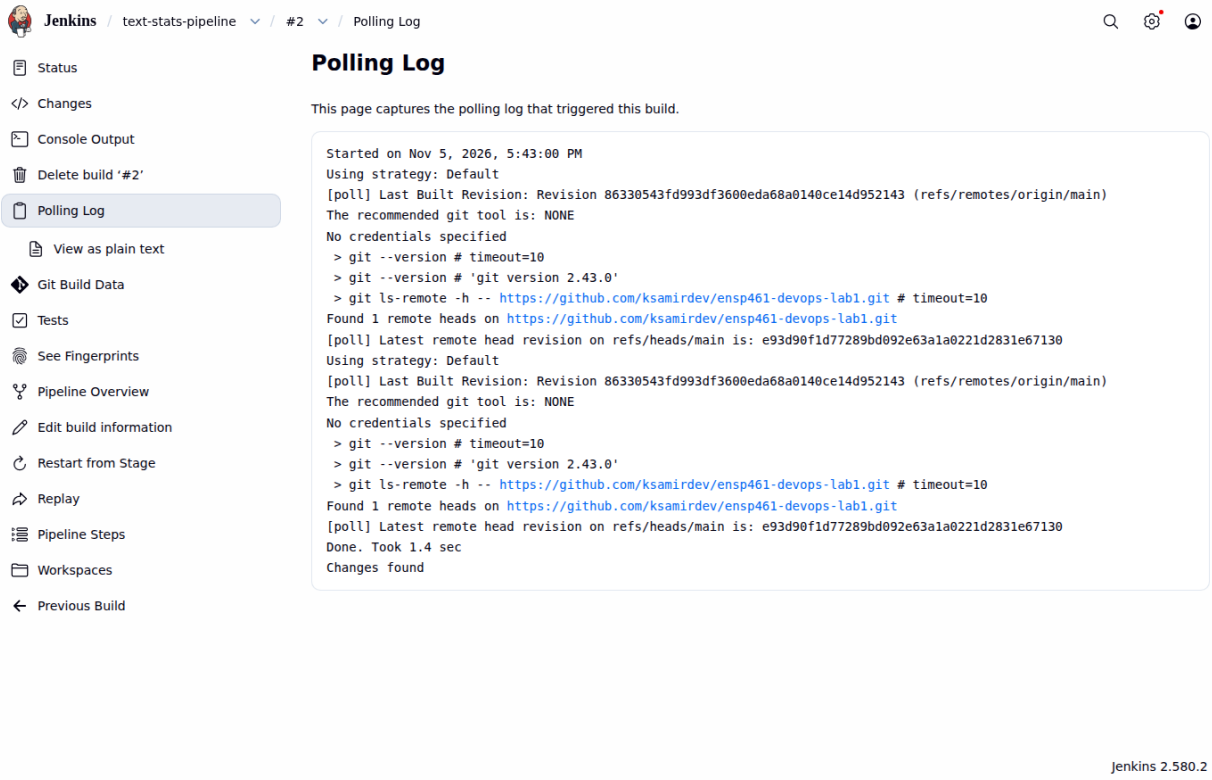


Fig: Polling log of build 2: change found on main

Post-build actions

The pipeline archives the tarball with fingerprints, publishes JUnit results, and prints a success or failure message in its post block.

Deployment steps

```
#!/bin/bash
# Installs a packaged build into /opt/text-stats, restarts the service and verifies health.
set -euo pipefail
archive="$1"
dest=/opt/text-stats

mkdir -p "$dest"
rm -rf "$dest/service"
tar xzf "$archive" -C "$dest"
[ -d "$dest/.venv" ] || python3 -m venv "$dest/.venv"
"$dest/.venv/bin/pip" install -q -r "$dest/requirements.txt"
systemctl restart text-stats

for _ in $(seq 1 15); do
    if body=$(curl -fsS http://localhost:5000/healthz); then
        echo "Health check passed: $body"
        exit 0
    fi
    sleep 1
done
echo "Health check failed" >&2
exit 1
```

```
[Unit]
Description=Text statistics service
After=network.target

[Service]
User=www-data
WorkingDirectory=/opt/text-stats
ExecStart=/opt/text-stats/.venv/bin/flask --app service.web run --host 0.0.0.0 --port 5000
```

```
Restart=on-failure

[Install]
WantedBy=multi-user.target
```

Automatic build and deployment

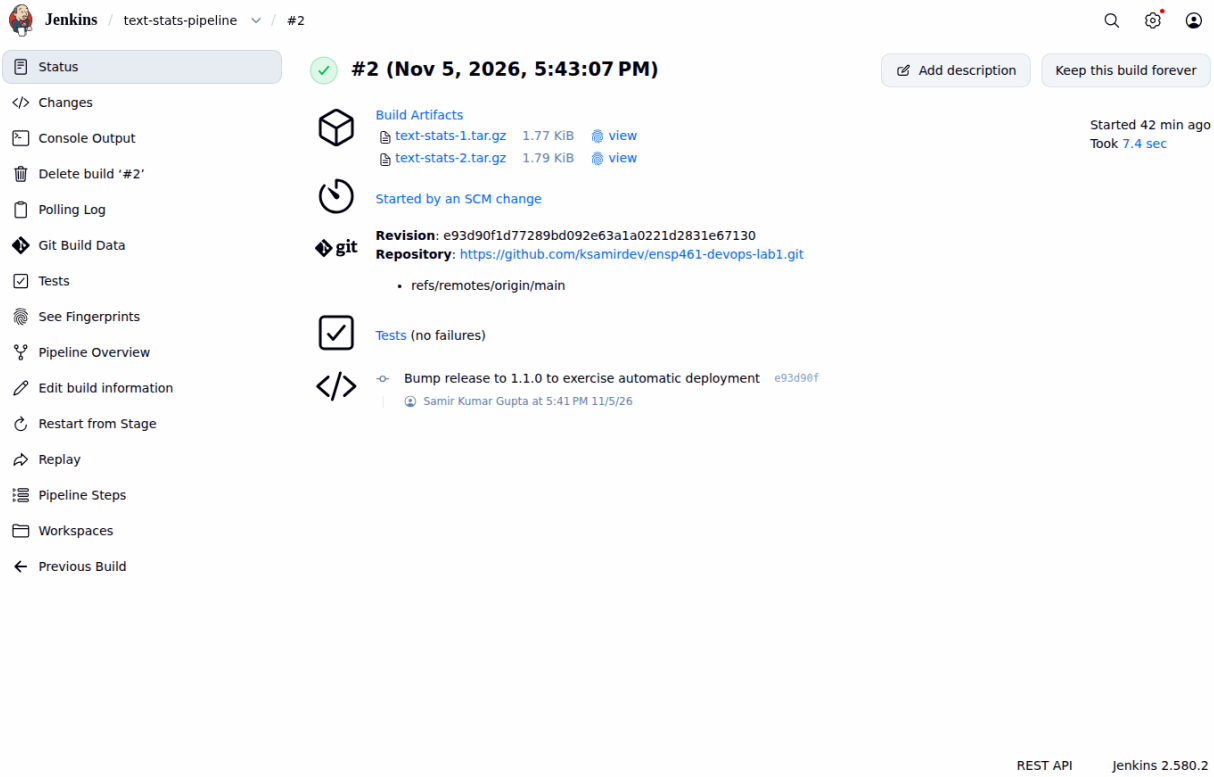


Fig: Build 2 started by an SCM change

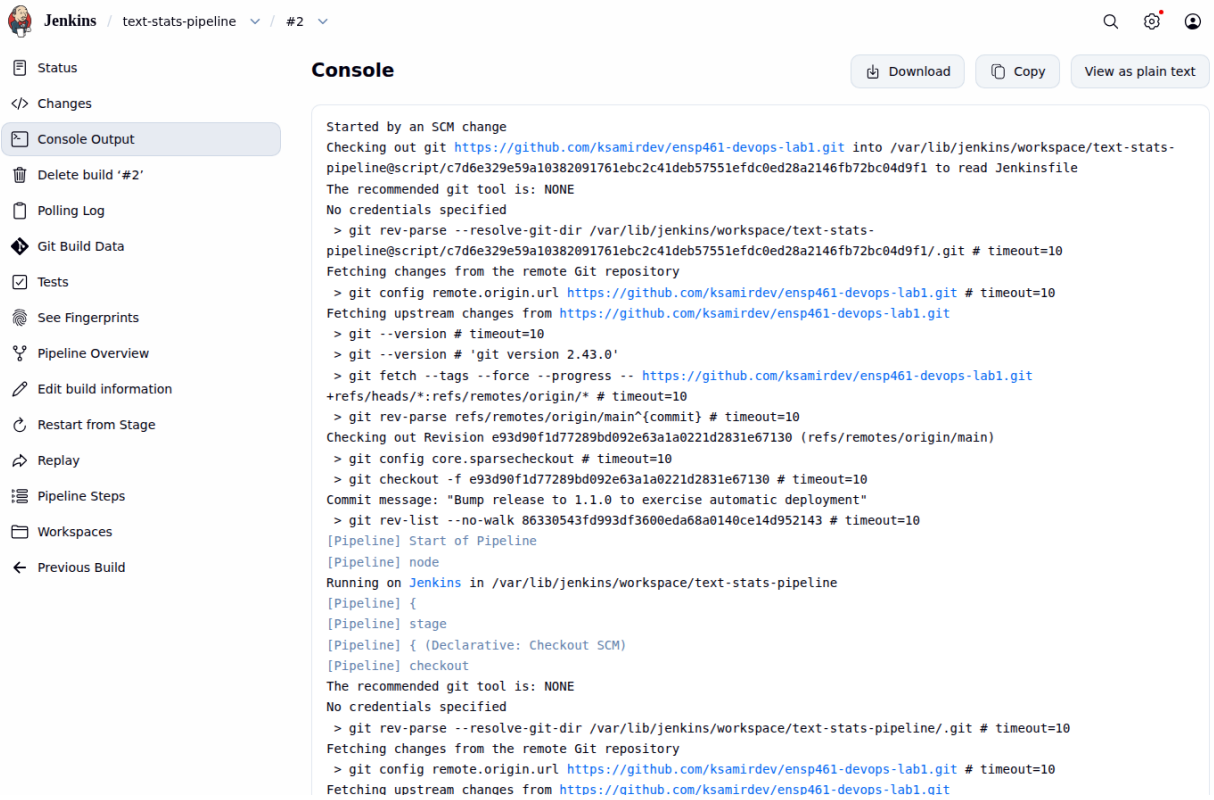


Fig: Build 2 console

```
$ systemctl is-active text-stats
active

$ curl -s http://localhost:5000/healthz
{"healthy":true,"release":"1.0.0"}

$ curl -s http://localhost:5000/
{"release":"1.0.0","service":"text-stats"}

$ curl -s -X POST -H 'Content-Type: application/json' -d '{"text":"build test build"}'
http://localhost:5000/stats
{"characters":16,"unique_words":2,"words":3}

$ curl -s -o /dev/null -w '%{http_code}\n' -X POST -H 'Content-Type: application/json' -d '{}'
http://localhost:5000/stats
400

$ curl -s http://localhost:5000/healthz
{"healthy":true,"release":"1.1.0"}

$ ls -l /opt/text-stats
total 4
-rw-r--r-- 1 jenkins jenkins 27 Nov  5 17:40 requirements.txt
drwxr-xr-x 1 jenkins jenkins 56 Nov  5 17:43 service
```

Fig: Service checks after builds 1 and 2 (rendered from evidence/logs/q3-deploy-verify.log)

Observations

- Build 2 was started by an SCM change, without clicking Build Now, and finished in about 7 seconds.
- After build 2 the /healthz endpoint reported release 1.1.0, confirming the new code went live.
- sudo for the jenkins user is limited to one root-owned script. That narrows the rule but is not a full security boundary, because the script still installs Python packages from the archive as root.
- One later poll took 58 seconds and logged a stack trace, with no effect on either build. The cause was not investigated, and later polls completed normally.